Digital Technologies in health and social care

Exploring the benefits and ethical concerns of digital technologies in practice

Introduction

- **Digital technology**: Any tool (software or hardware) that makes use of computers and the internet
- **Today's aim**: Wide breadth — not deep expertise in each, but critical awareness of all
- **The recurring question**: Who benefits? Who is excluded? Who is accountable?

The ethical challenge is rarely the technology itself — it is the assumptions built into it, and the conditions under which it is deployed.

SECTION

Evidence and new teams

Online resources and algorithmic gatekeeping

Digital support networks and the Quantified Self

Robotics and human connection

Evidence-based practice

Online databases and search engines have democratised access to research. The same evidence is available to a practitioner in a rural team as to one in a teaching hospital.

- **Algorithmic gatekeeping**: Search results are curated for engagement, not accuracy. What the platform surfaces is not necessarily what is most rigorous.
- **Data-driven insights**: Population-level data identifies trends and targets interventions, but risks turning individuals into data points rather than people.

Evidence-based practice (2)

- **Source quality**: Not all online evidence is equal. Practitioners need to distinguish peer-reviewed research from opinion and recognise commercial interests.
- **Personal information management**: Digital tools for organising patient data support practice, but raise questions about privacy, consent, and access.
- **Critical inquiry**: Digital tools work best alongside professional curiosity, not instead of it.

> If Google shapes what counts as evidence, who is really determining best practice?

New healthcare teams (1)

- **Social media**: Peer support at scale for isolating conditions; misinformation spreads faster than correction, and boundaries blur online.
- **Quantified self**: Self-monitoring via wearables supports engagement in health, but excludes those who can't afford devices, creating new anxieties.
- **Citizen science**: Public participation validates lived experience and generates data at scale, though it challenges credentialed expertise in ways requiring navigation.

> A service user arrives having extensively researched their condition online. What is your professional role in that conversation?

New healthcare teams (2)

Robotics in Care

- **Assistive robots**: Exoskeletons and lifting aids reduce physical strain and deliver consistent rehabilitation support.
- **Social companions**: PARO — a robotic seal used with dementia patients — provides comfort and stimulation.
- **Unintended consequence**: Patients form real emotional attachments to PARO. It responds — but does not understand. Is that connection meaningful, or the appearance of care at the point it matters most?

The line between *augmenting* and *substituting* is rarely drawn by the people receiving care.

For discussion

1. What does it mean for a profession when its evidence base is partly curated by commercial algorithms?
2. Where is the line between a service user being "informed" and being misled by online health information — and whose responsibility is it?
3. If a robot can provide consistent companionship to a dementia patient, does the authenticity of the connection matter?
4. How do you balance respecting a service user's self-gathered knowledge with your own professional responsibility?

SECTION

Population health

The Digital First agenda and the Digital Divide

Digital "nudges" and behaviour change

Genomics and personalised medicine

Population health (1)

- **Digital first**: Policy agenda to make digital the default route for services — GP appointments, benefit claims, social care referrals
  - Reduces cost and increases availability — but assumes digital access and confidence many vulnerable service users do not have
- **The Digital Divide**: Not just about device ownership
  - Also about confidence, trust, and whether systems were designed with people like them in mind
  - Those most in need of services are often most excluded by digital delivery

Population health (2)

- **Nudges**: Subtle interface choices that steer behaviour without prohibiting alternatives — opt-in defaults, push notifications, calorie prompts
  - Effective at scale — but whose definition of "healthy" is encoded in the design?
- **Genomics**: Genetic data used to personalise treatment and predict risk before symptoms appear
  - Raises risks of genetic discrimination and the "burden of knowledge" for untreatable variants

> If a service is "digital first," what does that mean for someone who is elderly, homeless, or has low digital literacy?

For discussion

1. Who decides what "healthy behaviour" looks like — and what happens when that definition gets encoded into a digital system used at scale?
2. At what point does a "digital first" agenda become a form of structural discrimination?
3. If genomic data can predict serious illness decades in advance, should everyone have access to that information? Who should hold it?
4. How would you advocate for a service user who is being failed by a system that assumes digital access they don't have?

SECTION

Artificial intelligence

The AI Spectrum: From rules to generation

When AI fails: Pulse oximetry and skin tone

Ethics: High-stakes decisions, metadata, and generative AI

The AI spectrum

"Artificial Intelligence" is not one thing. Understanding the spectrum helps us identify where the risks lie.

Type	What it does	Example	Key risk
Rule-based	Follows "If-Then" rules	Prescribing alerts	Rules reflect the biases of whoever wrote them
Machine Learning	Finds patterns in data	Sepsis risk scores	Historical data encodes historical inequalities
Deep Learning	Analyses complex signals	Radiology screening	Hard to explain why a decision was made
Generative AI	Creates new content	Chatbots, case notes	Plausible outputs that may be fabricated

Wartmann & Combs (2018)

When AI fails

Algorithmic bias is not a glitch; it is a systematic failure that often maps onto existing social inequalities.

- **Pulse Oximetry**: Devices used in COVID-19 overestimated blood oxygen in Black patients because the algorithm didn't account for skin tone. Result: Delayed treatment (Sjoding et al., 2020).
- **Dermatology AI**: Diagnostic tools trained on 95% lighter skin tones perform significantly worse on darker skin (Daneshjou et al., 2022).
- **Automation Bias**: Trusting a computer recommendation even when it contradicts professional judgement.

Ethics

- **High-stakes decisions**: AI informs child protection referrals, benefit eligibility, and criminal risk scores. A "human in the loop" can become a rubber stamp, approving without time to challenge.
- **Metadata**: Timestamps, location, who you contacted, can reveal details without accessing content; rarely collected with meaningful consent.
- **Generative AI**: Drafts case notes and referral letters with fluency. The danger: outputs that sound right, even when fabricated.

> An AI tool flags a family as "high risk" for child neglect. You disagree. What do you do — and who is responsible for the outcome?

For discussion

1. How confident would you need to be in your own judgement to override an AI recommendation in a high-stakes decision?
2. If an AI tool is biased against certain groups but still performs better on average than human assessors, what do you do?
3. What does "informed consent" mean when a patient cannot — realistically — understand how their data is being used?
4. Who is accountable when a generative AI produces a fabricated case note that influences a decision about a vulnerable person?

SECTION

Consultation and your role

Records, VR, Telehealth, and Chatbots

Leadership in a digital world

Moving from "Operator" to "Advocate"

Consultation & support (1)

- **Electronic health records**: Enable continuity across services, but practitioners who read the record first arrive with a pre-formed narrative. "The notes become the patient."
  - Fragmented systems can hinder rather than help information sharing
- **Virtual reality**: Immersive environments for phobia exposure, pain management, and trauma processing — controllable and repeatable
  - Risk: desensitisation, and confusion about the boundary between virtual and real
- **Telehealth**: Removes geography as a barrier and reduces stigma around help-seeking
  - Non-verbal cues are lost on screen, and being in someone's home shifts the dynamic unpredictably

Consultation & support (2)

- **Chatbots**: Available at 3am when crisis lines are busy; a low-stigma entry point for those not ready to speak to a person
  - Cannot read genuine distress — someone in crisis may feel more alone after a scripted response
- **Digital triage tools**: Symptom checkers and risk assessment apps aim to direct people to the right level of care
  - Risk: over- or under-triage, and reinforcement of existing inequalities

> Can you build a genuine therapeutic alliance through a screen? Does it depend on the person, the problem, or the platform?

For discussion

1. What is lost — and what might genuinely be gained — when a therapeutic relationship moves online?
2. How do you avoid letting a detailed case record replace your own direct assessment of who someone is?
3. At what point does a chatbot providing mental health support become an ethical problem rather than a useful resource?
4. How do you decide when to recommend a digital tool and when to insist on a human alternative?

SECTION

Leadership

Digital competence as a professional skill

Advocacy and the digital divide

Innovation, solitude, and identity

Leadership

- **Digital competence**: Asking who built this, for whom, on what data. Can't interrogate it? You're at the mercy of its designers.
- **Advocacy**: Digital tools can amplify marginalised voices, but those who need advocacy most are least likely to reach it.
- **Innovation**: "We use an AI risk tool" ≠ "our tool is fair and effective." Resist the pressure to adopt first, evaluate later.
- **Solitude**: Quiet, uninterrupted reflection is where professional judgement forms — and increasingly hard to protect.

> What is the difference between a practitioner who *uses* digital tools and one who *understands* them? Does that distinction matter for your

For discussion

1. What would "digital competence" look like as a formal requirement for HSC practitioners — and who would assess it?
2. How do you evaluate whether a digital innovation is genuinely improving care, versus improving the *perception* of care?
3. In a culture of constant connectivity, how do you protect the time and space needed for reflective professional practice?
4. What does ethical leadership look like when the tools your organisation uses may be causing harm to some of the people it serves?

Your role as a professional

You will not just "operate" technology; you will work **alongside** it. Your role shifts to:

- **The Interpreter**: Translating digital outputs into decisions that account for context the record doesn't capture.
- **The Advocate**: Making the case for a human alternative when someone is being excluded by a digital-first system.
- **The Critical Enquirer**: Knowing the limitations of algorithms — and having the courage to override them when your judgement says they're wrong.

Technology does not replace professional responsibility. In many cases, it raises the stakes.

Conclusion

Digital technology is a powerful tool for **efficiency**, but it is not a substitute for **professional judgement**.

As social workers, you are the guardians of the human element in care.

Through thoughtful advocacy and critical evaluation, we can harness technology to improve lives without sacrificing the **human connection** that defines our profession.